

Program 10 : Demonstrate Kubernetes Dashboard and CLI for clustering monitoring.

Step1: Initiate minikube

```
>minikube start --cpus=2 --memory=4096
```

Step2: Enable Kubernetes dashboard

```
> minikube dashboard
```

Step3: create a Deployment file

```
apiVersion: apps/v1
```

```
Kind: Deployment
```

```
Metadata:
```

```
Name: nginx-deployment
```

```
Labels:
```

```
app: nginx
```

```
spec:
```

```
Replicas: 2
```

```
selector:
```

```
Matchlabels:
```

```
App: nginx
```

```
spec:
```

```
Containers:
```

```
-name: nginx
```

```
Image: nginx:latest
```

```
Ports:
```

```
-containerPort: 80
```

Step4: Create a Service file

```
apiVersion: v1
```

```
Kind: service
```

```
metadata:
```

```
name: nginx-service
```

```
spec:
```

```
App: nginx
```

```
Ports:
```

```
-protocol: TCP
```

```
Port: 80
```

```
targetPort: 80
```

```
Type: nodeport
```

Step5: Apply Resources

```
>kubectl apply -f deployment.yaml
```

```
>kubectl apply -f service.yaml
```

Step6: Monitor via CLI

> kubectl get pods

Step7: check service

>kubectl get svc

Step8: check pods logs

>kubectl logs <pod-name>

Step9: describe pods

> kubectl describe pod <pod-name>

Step10: Access the application ia minikube and viewing the same in the web browser

> minikube service nginx service

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:32675

🌈 Opening service default/nginx-deployment in default browser...



Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

